

Two-Seed Brightness Experiment

A plain-language record of the completed real-only versus synthetic brightness comparison

What was tested

A compact U-Net trained on Rural LoveDA imagery at 512 x 512 pixels. One model used only real images. The other used a final training mix containing 10% brightness-adjusted copies.

How it was checked

Each setup was trained twice, with random seeds 42 and 7. Frozen checkpoints were evaluated on the same Rural validation masks under five controlled brightness conditions.

Headline finding

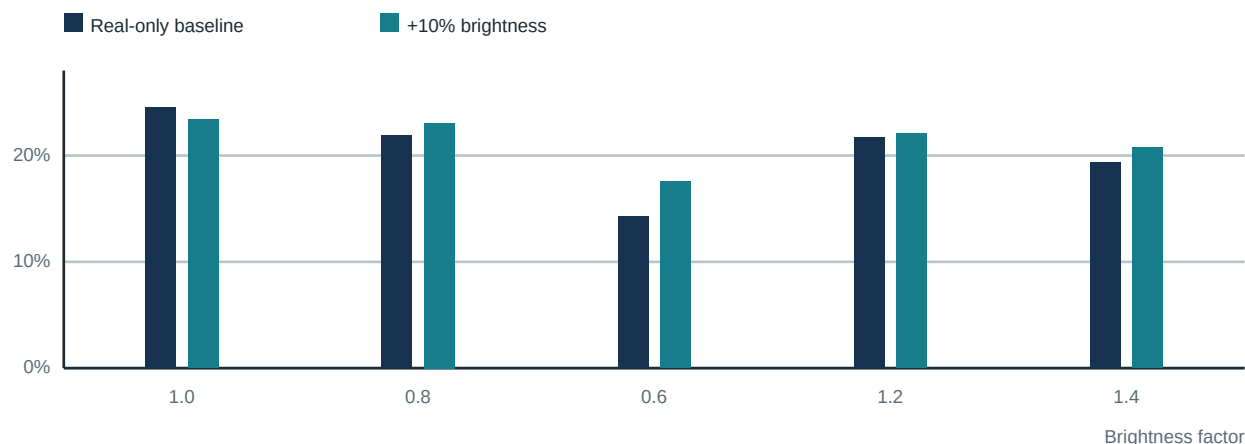
Brightness augmentation reduced clean-image mIoU by 1.13 percentage points on average, but improved mIoU under every tested lighting shift. The largest average benefit was +3.29 points for much darker images (brightness 0.6).

Experiment settings held constant

Setting	Value
Training data	LoveDA Train/Rural: 1,366 real image/mask pairs
Validation data	LoveDA Val/Rural: 992 image/mask pairs
Model	Compact U-Net, 7 classes, base channels 16
Training	15 epochs, Adam, learning rate 1e-4, batch size 1
Synthetic change	10% final mix, brightness factors 0.8 and 1.2; labels unchanged
Metrics	mIoU, class IoU, and pixel accuracy; No Data ignored

Average Robustness Results

Mean validation mIoU across seeds 42 and 7. Higher is better. Percentage-point changes compare the 10% brightness model to the real-only model.



Brightness condition	Real-only	+10% brightness	Change
Clean (1.0)	24.54%	23.41%	-1.13%
Darker (0.8)	21.88%	23.07%	+1.19%
Much darker (0.6)	14.24%	17.53%	+3.29%
Brighter (1.2)	21.67%	22.06%	+0.39%
Much brighter (1.4)	19.37%	20.77%	+1.40%

How to read this result

The real-only model is the better choice if expected inputs are clean and match the original dataset. The brightness-trained model is more dependable when images may be notably darker or brighter than the training imagery. This is a tradeoff, not an overall winner.

The brightness model improved under all four non-clean test conditions in the two-seed average. The largest and most useful gain occurred at brightness 0.6, a deliberately difficult dark-image condition.

Interpretation and Next Step

What the completed work establishes, what it does not yet establish, and how to strengthen the evidence.

What is supported

With the tested compact U-Net and this specific augmentation recipe, 10% mild brightness synthetic data consistently traded a small amount of clean validation performance for better performance when image brightness changed. Both seeds show this direction under strong darkening.

Important limitations

This result currently comes from two random seeds, one model size, one dataset region, one synthetic ratio, and one augmentation type. It does not prove that brightness augmentation will always help or that 10% is optimal. The Barren class also remained at 0 IoU, so this experiment did not solve class imbalance.

Recommended next step

Run one final paired experiment with seed 123 using the exact same settings. Then average all three paired results. If the same robustness pattern remains, record 10% brightness augmentation as a supported robustness intervention and move on to a different synthetic-data question.

Files used for this record

evaluation/brightness_robustness/results.json (seed 42) and evaluation/brightness_robustness_seed7/results.json (seed 7).